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char type

Table 9.7 Memory allocation for char Data types

Data type		Space in memory		Range of value
		in terms of bytes	in terms of bits	
char	Signed ASCII character	1 byte	8 bits	-128 to 127
unsigned char	ASCII character without sign	1 byte	8 bits	0 to 255
signed char	ASCII character with sign	1 byte	8 bits	-128 to 127

Floating point type

Table 9.8 Memory allocation for floating point Data types

Data type		Space in memory		Range of value
		in terms of bytes	in terms of bits	
float	signed fractional value	4 bytes	32 bits	3.4×10^{-38} to 3.4×10^{38}-1
double	signed more precision fractional value	8 bytes	64 bits	1.7×10^{-308} to 1.7×10^{308} -1
long double	signed more precision fractional value	10 bytes	80 bits	3.4×10^{-4932} to 1.1×10^{4932} -1

Memory allocation is subjected to vary based on the type of compiler that is being used. Here, the given values are as per the **Turbo C++** compiler. **Dev C++** provides some more space to **int** and **long** double types. Following Tables 9.9 shows the difference between Turbo C++ and Dev C++ allocation of memory.

Table 9.9 Memory allocation by Turbo C++ and Dev C++

Data type	Memory size in bytes	
	Turbo C++	Dev C++
short	2	2
unsigned short	2	2
signed short	2	2
int	2	**4**
unsigned int	2	**4**
signed int	2	**4**
long	4	4